Royal Education Society's

College of Computer Science and Information Technology, Latur.

Affiliated to

Swami Ramanand Teerth Marathwada University, Nanded.

A Project Synopsis

On

Weather Update Web-App (OpenWeather API)

Submitted for the award of degree of

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Project Guide

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Abstract –

This project aims to develop a **Real-Time Weather Forecasting Web Application** using the **OpenWeather API**. The application enables users to retrieve accurate and up-to-date weather reports for any city or location in the world. Built using **HTML, CSS, JavaScript and** along with weather-related icons and images, the system provides a simple yet efficient interface for real-time weather data access. The traditional way of checking weather via TV or multiple websites is often time-consuming and scattered. This web app solves that issue by offering a centralized and user-friendly platform that displays real-time weather conditions such as temperature, humidity, wind speed, wind direction, and forecast data. This project helps to get Real-Time Weather information from any city and location in the world.

Introduction:

The **Real-Time Weather Forecasting Web App** is designed to simplify the way users check weather updates by integrating the **OpenWeather API** with modern web technologies. The system displays weather information instantly based on the user's input or current location, removing the need to visit multiple sources or manually interpret data.

The main objectives of the project are:

- To provide **real-time and location-based weather updates**.
- To implement a clean and responsive **user interface using HTML, CSS and JavaScript**
- To fetch weather data such as **temperature, humidity, wind speed, and weather descriptions** using the **OpenWeather API**.
- To make weather information accessible, accurate, and user-friendly through visual elements like **icons and images**.
- To offer an educational resource for those learning **API integration and front-end web development**.

This web application will assist users in planning their daily activities based on current and forecasted weather.

Project Module:

1. Home Module

The **Home Module** is the main interface of the Real-Time Weather Forecasting Web App. It displays current weather and a 5-day forecast using data from the **OpenWeather API**.

Key Features:

- **City Search:** Users can enter a city name (e.g., *Latur*) to view live weather updates.
- **Current Weather:** Shows temperature, condition (e.g., *broken clouds*), and a weather icon.
- **Additional Info:** Includes air quality, real feel, humidity, temperature range, wind speed/direction, visibility, pressure, sunrise & sunset times.
- **5-Day Forecast:** Displays daily temperature, condition (e.g., *Rain*, *Clouds*), and icons.
- **Design:** Clean UI with weather-themed background, responsive layout, and icons.

Tech Stack:

- **Frontend:** HTML, CSS and JavaScript
- **Backend:** JavaScript
- **API:** OpenWeather API
- **Assets:** Weather icons, background images

Project Plan:

Gantt chart:

Sr.no	Task Name	1-Aug	1-Sep	1-Nov	1-Dec	25-Feb
1	Requirement Gathering					
2	Planning					
3	Designing					
4	Coding					
5	Testing and Deployment					

Project Requirements:

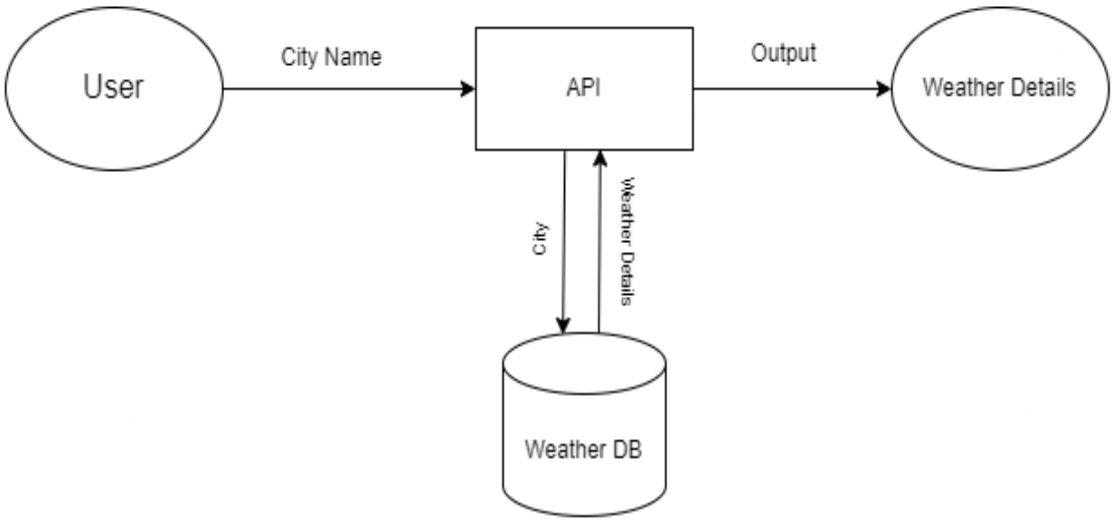
➢ Hardware Requirements:

1) Processor: i5 and above
2) RAM: 4GB and above
3) Hard Disk: 256 GB or above
4) Input device: keyboard, Mouse
5) Output device : Monitor or LCD/LED, Printer
6) LAN Network or Internet Connection

➢ Software Requirement:

- Frontend-IDE: Visual studio 2008 or above using ASP.NET
- Language : HTML, CSS and JavaScript
- Framework: -
- Backend-Database: -

Data Flow Diagram:



Conclusion:

Through this project, we successfully developed a **Real-Time Weather Update Web App** using **HTML, CSS and JavaScript** and the **OpenWeather API**. The application allows users to search and view current weather information for any city in the world, with a clean and interactive user interface.

Working on this project gave us valuable hands-on experience with **API integration**, **frontend and backend web development**, and building a **full-stack web application**. We also learned how to design responsive layouts and handle real-time data dynamically.

Overall, this project helped us improve our technical skills and gave us insight into real-world web application development. In the future, we plan to enhance the app by adding features like weather alerts, favourite cities, and support for multiple languages.

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